Bare Bones Arduino is a $3 board that leaves little explanation necessary after its name. It is basically the Atmega chip used on the Arduino, which can be used this way because the Arduino actually is just a programming environment for the actual Atmega microcontroller. At 3$, the performance drops hardly matter.

mbed

A collaborative software hardware development platform maintained by ARM, the hardware side comprises of the mbed Microcontroller board based on an NXP microcontroller, which has an ARM Cortex M3 core, running at 96 MHz, with 512 KB flash, 64 KB RAM, as well as several interfaces including Ethernet, USB Device, CAN, SPI, I²C and other I/O

Pinguino

More of an add-on to the original Arduino aiming to enhance its capabilities, this chip is not manufactured and most users build their versions themselves. The IDE is highly supportive and is adding more android compatible features as we speak.

Alongside this list, Teensy, Beagleboard, Nanode and TI Launchpad deserve special mentions as worthy options.

With this exhaustive list of alternatives, when you choose a certain piece the best way to check if it suits your needs is to check projects online and see it working.

NOTE: all prices mentioned so far are subject to variation. That's open source for you.

Print away

Most of the boards, and any semiconductor device that we see, are PCB, or Printed Circuit Board devices. On a PCB, the conductive channels are printed onto a non-conductive base to connect components, with single, double and multi-layer PCB's also possible and being made. There are two topics of interest to the maker community regarding PCB:

PCB Simulation

Simulating a PCB Design can be crucial in your choice of a board, with several softwares providing you templates to simulate popular boards like Arduino and the Pi. It is also a very crucial step in the process of designing a chip or board by providing realistic testing data without the price over-